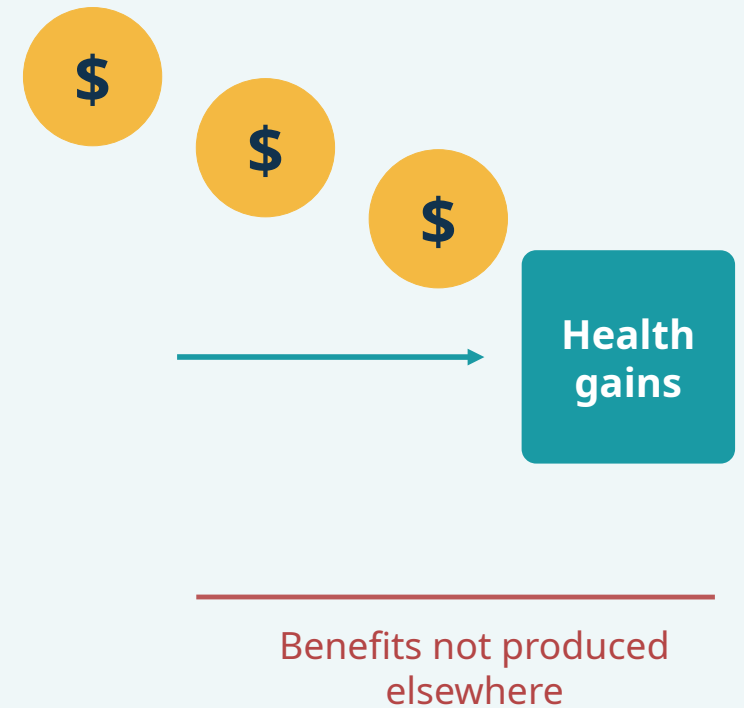


Basic Economic Evaluations

Core concepts for cost-effectiveness and cost-utility analysis

Key theme: economic evaluation does not ask “what is cheapest?”
It asks which option makes the best use of scarce resources.



Why economic evaluation?

Healthcare choices are trade-offs under scarcity.



The “cost” is not only money spent. It is also the health that could have been generated by using the same resources elsewhere.

Economic evaluation = comparative analysis of alternatives by both costs and consequences

What makes an evaluation “full”?

Two defining features: compare alternatives and consider both costs and consequences.

		No	Compares alternatives?	Yes
Considers both costs + outcomes?	No	Description or costing only		Cost comparison only
	Yes	Outcome study only		Full economic evaluation

EXAMPLES OF MEANINGFUL ALTERNATIVES

- New medicine vs standard treatment
- Screening program vs usual care
- Community care vs hospital care

Value for money requires both sides of the ledger

The major forms of economic evaluation

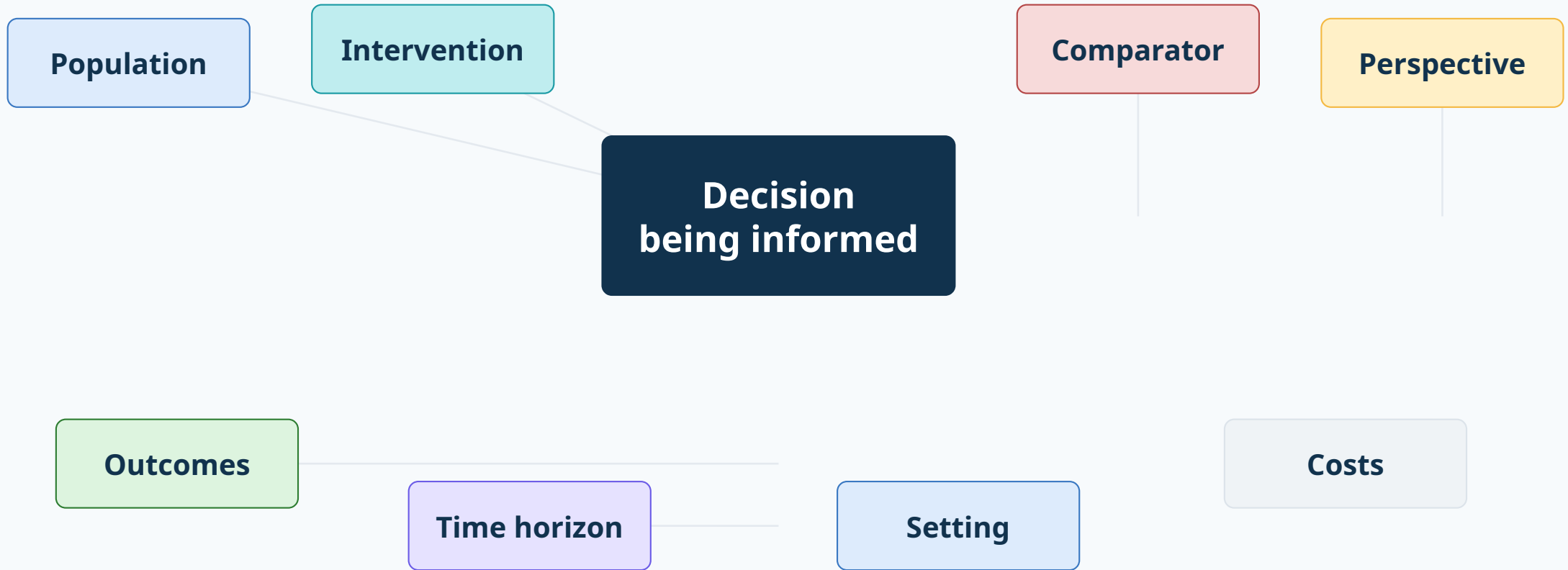
The cost side is usually monetary; the consequence side changes.

Method	Costs	Consequences	Typical result
Cost-effectiveness analysis	Costs in dollars	Natural health units	Cost per hospitalization avoided
Cost-utility analysis	Costs in dollars	QALYs or DALYs	Cost per QALY gained
Cost-benefit analysis	Costs in dollars	Benefits in dollars	Benefit-cost ratio or net benefit
Cost-minimization analysis	Costs in dollars	Outcomes assumed equivalent	Least-cost option

In healthcare, CEA and CUA are the main classroom focus.

Start with a decision problem

Define the question before building the spreadsheet or model.

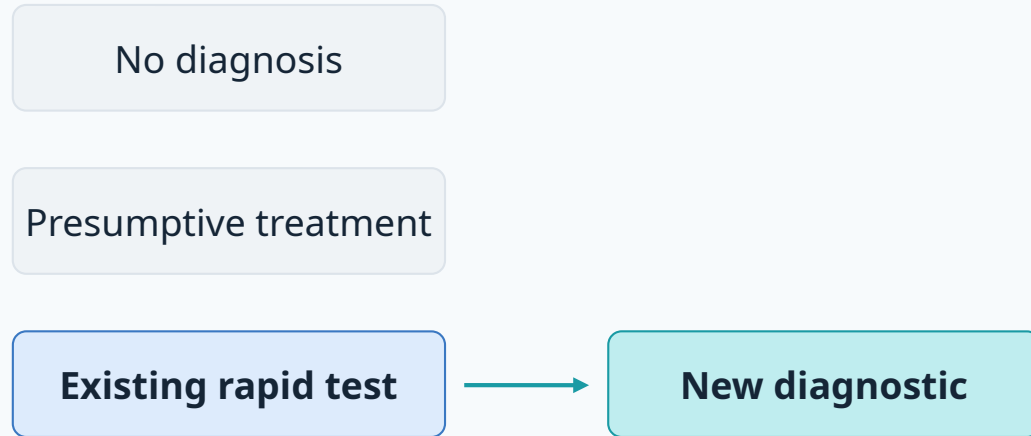


A good research question resembles PICO, but expands it to include costs, perspective, time, and decision context.

Comparator and time horizon can change the answer

Compare with realistic practice and follow costs and benefits long enough.

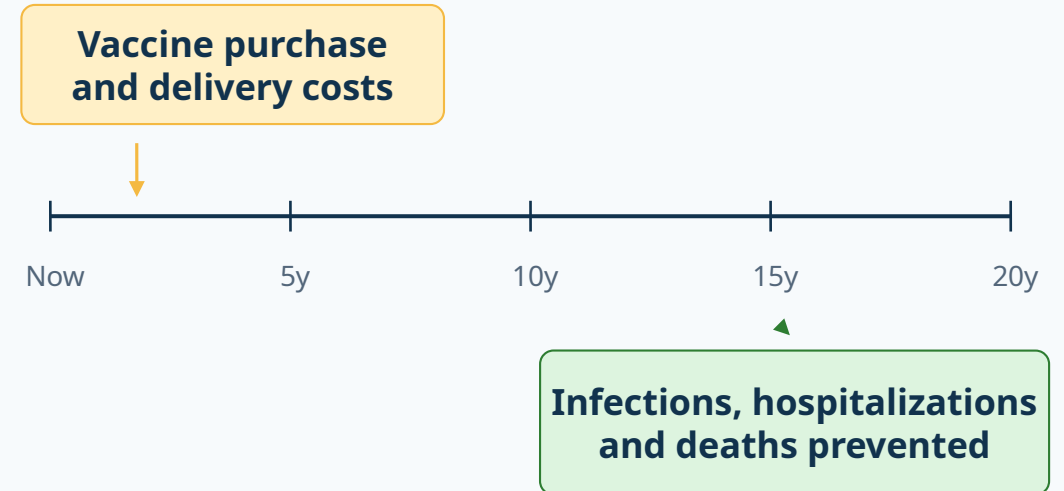
COMPARATOR CHOICE



The relevant question is rarely “better than nothing.”

Using an obsolete comparator can make an intervention look artificially attractive.

TIME HORIZON



Too short a horizon misses downstream outcomes and distorts prevention.

Perspective decides whose costs count

The same intervention can look different depending on the viewpoint.

Payer

Claims or program expenditures

Healthcare system

Medicines, tests, staff, visits, admissions

Patient + family

Transport, fees, time, caregiving

Society

All costs and consequences regardless of who bears them

Example: home-based treatment

**Health service
spends more**



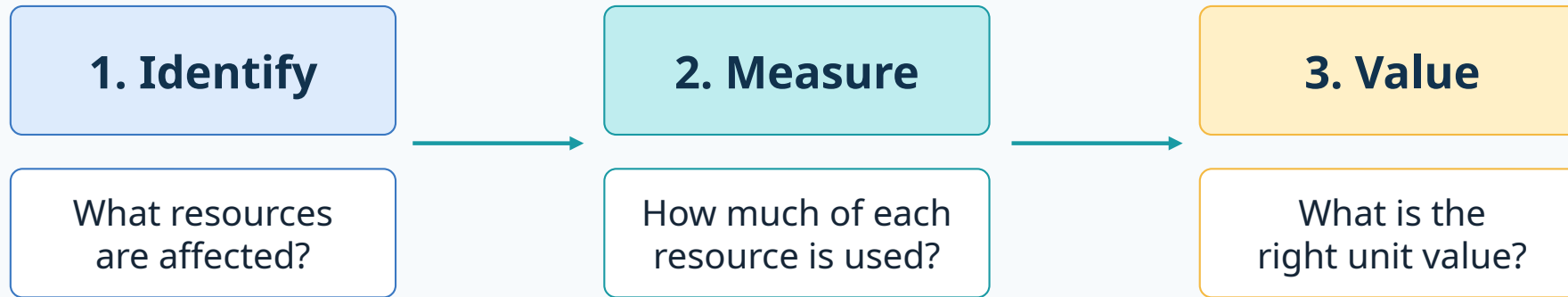
**Patients lose
less time + travel**



**Result changes
with perspective**

Costing: identify → measure → value

Cost estimation is a resource-use exercise, not just an accounting exercise.



RESOURCE EXAMPLES

Staff time • medicines • labs • buildings • equipment • utilities • transport • patient time • informal care

Economic cost exists when a resource has an alternative use — even if no payment occurs.

Consequences: measure real health effects

Best available clinical evidence still has to match real-world delivery.

EFFICACY VS EFFECTIVENESS



Adherence, stock-outs, provider behavior, and access can reduce realized value.

OUTCOME CHAIN

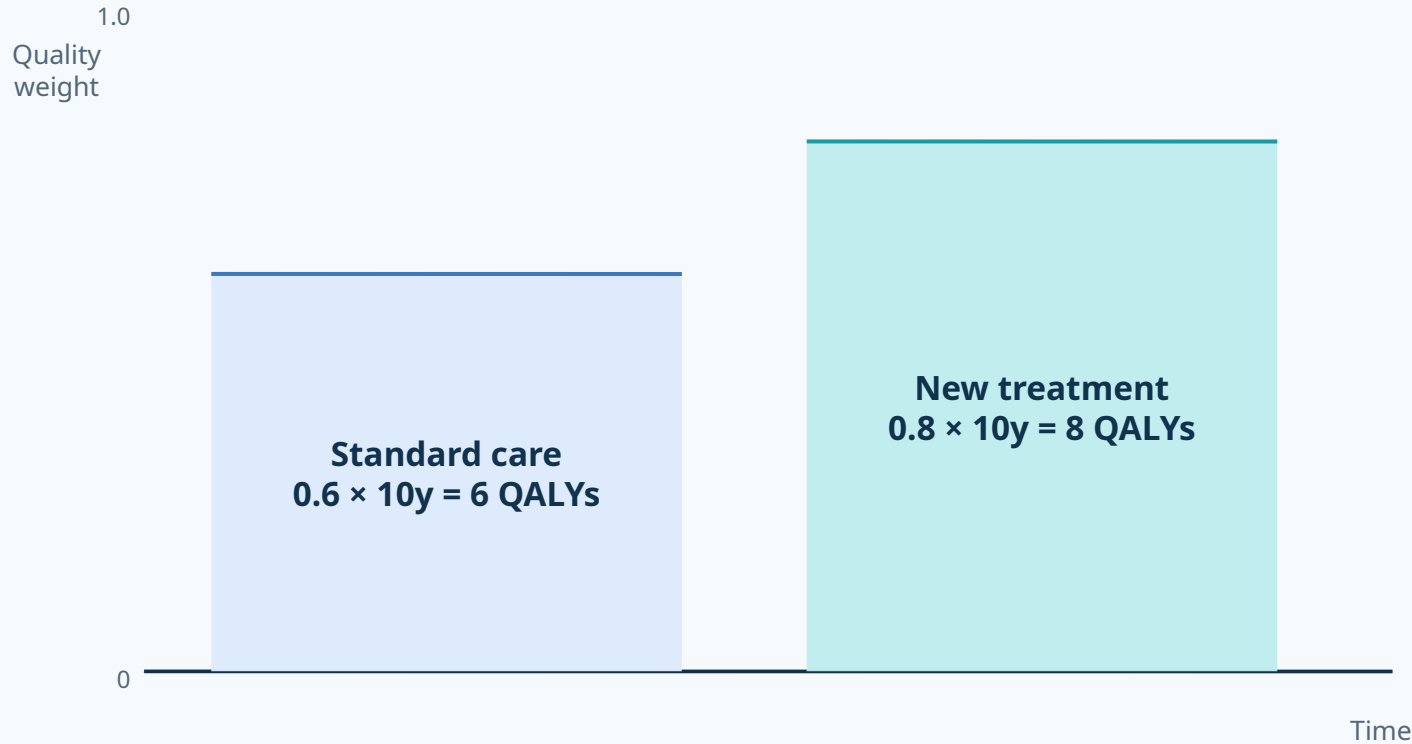


Intermediate outcomes are useful only when linked credibly to final health outcomes.

Evidence sources: randomized trials, observational studies, systematic reviews, routine data, and decision models.

QALYs combine length and quality of life

One QALY = one year lived in full health.



Example: arthritis treatment improves quality of life, not survival.

$$\text{QALYs} = \sum (\text{utility weight} \times \text{time})$$

$$\Delta\text{QALYs} = 8 - 6 = 2$$

- Captures survival gains
- Captures quality-of-life gains
- Supports comparisons across diseases

CEA and CUA answer related but different questions

Natural units are tangible; QALYs enable cross-program comparison.

Cost-effectiveness analysis

Outcomes in natural health units

infection
prevented

hospitalization
avoided

life-year
gained

Easy to understand inside one clinical area

Limited for cross-disease league tables

Cost-utility analysis

Outcomes in QALYs or DALYs

health
profile



quality × time



QALYs

Allows comparison across clinical areas

May miss broader social values and equity concerns

Think incrementally: the ICER

Decision-makers ask what the additional health gain costs.

$$\text{ICER} = \Delta\text{Cost} / \Delta\text{Effect}$$

Where Δ means "new option minus comparator."

Option	Cost	QALYs
Standard	\$1,000	3.0
New	\$1,600	3.2

$$\Delta\text{Cost} = \$600$$

$$\Delta\text{Effect} = 0.2 \text{ QALYs}$$

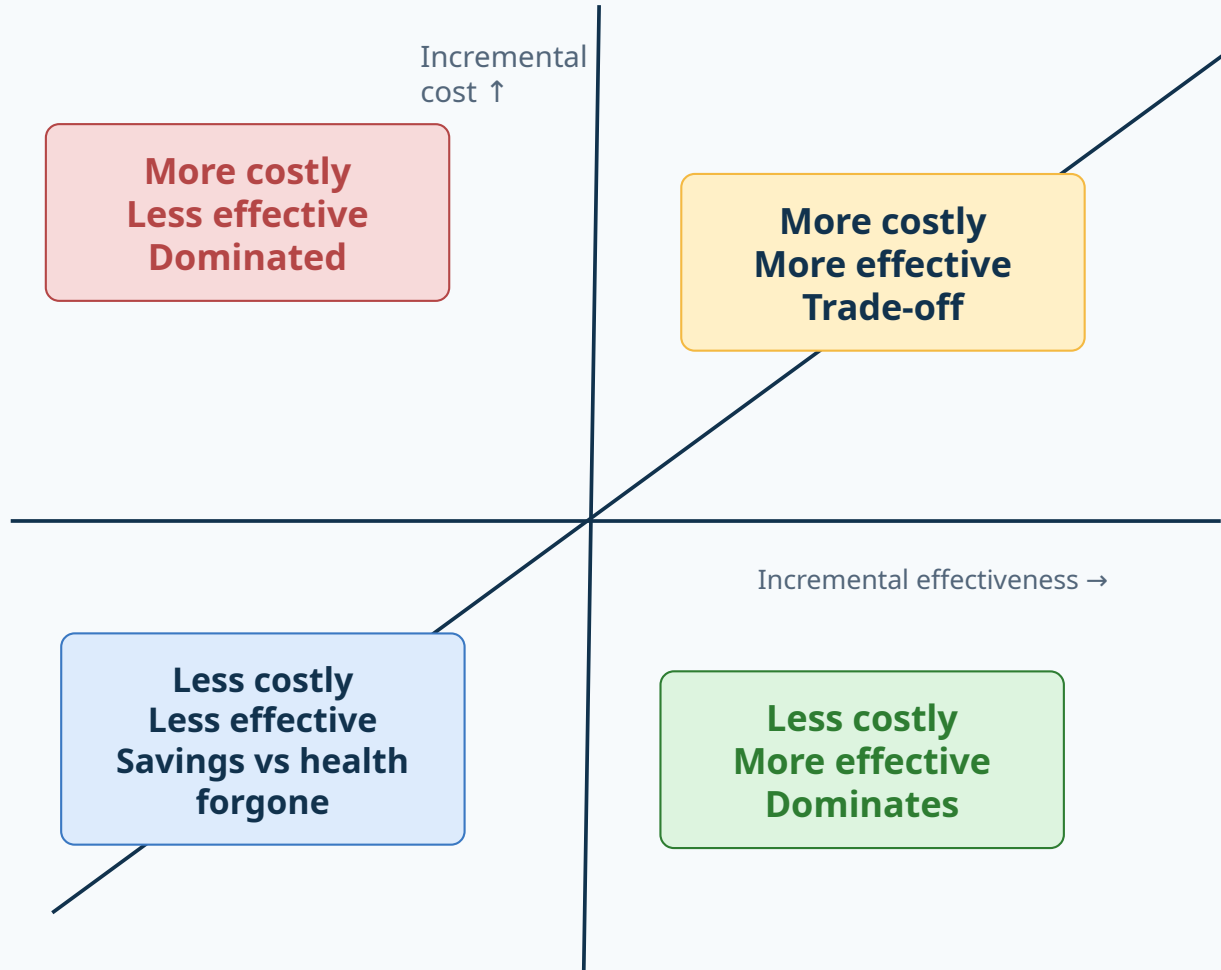


$$\$600 / 0.2 = \$3,000 \text{ per QALY}$$

Average cost-effectiveness ratio answers a different question; the decision is about the increment.

Cost-effectiveness plane: four decision regions

A graphical way to interpret incremental cost and effect.



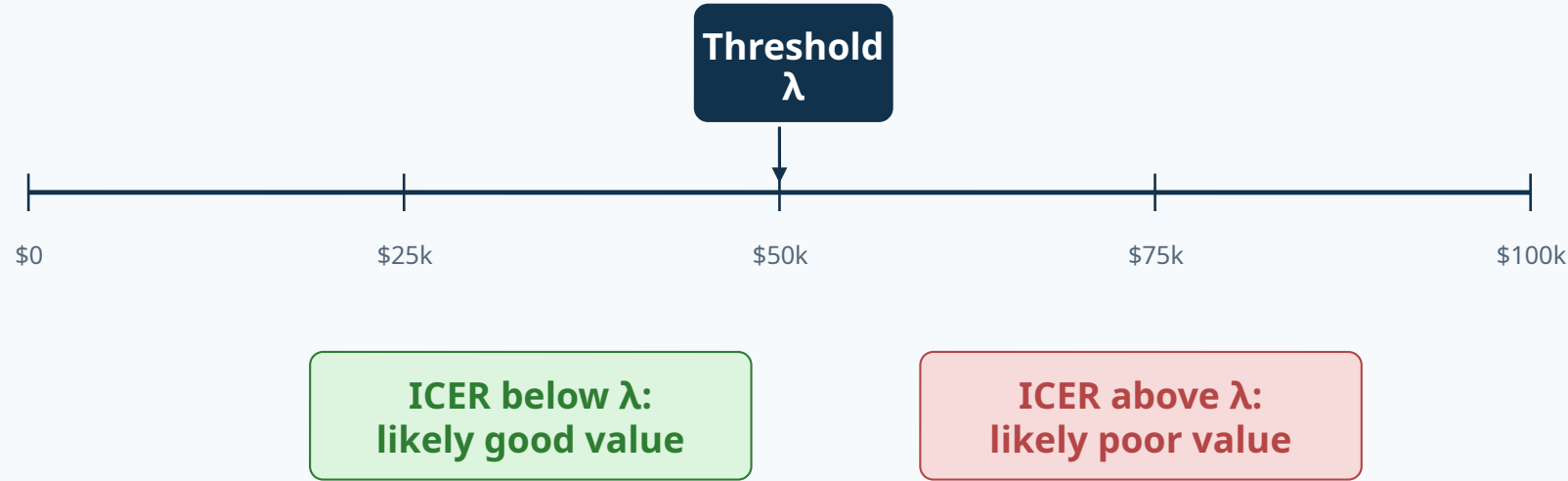
DOMINANCE RULES

- Remove options that cost more and produce less.
- Rank multiple options from least to most effective.
- Recalculate ICERs after removing dominated options.

Do not compare every option only with a common baseline.

Thresholds express opportunity cost

When an intervention is better and more costly, compare the ICER with a threshold.



The deeper question

Health gained
by new program

vs

Health displaced
elsewhere

Thresholds are not universal scientific constants. They vary with budgets, prices, disease burden, and health-system opportunity costs.

Cost-effective $\neq$ cost-saving $\neq$ affordable

These terms answer different questions.



DECISION TRAPS

- Low price does not necessarily mean good value.
- Good value per patient can still create budget pressure at scale.
- Cost-effective interventions often increase spending but produce worthwhile health gains.
- Budget-impact analysis complements CEA when annual spending may change materially.

Discounting changes how prevention looks

Future costs and health consequences are converted into present values.

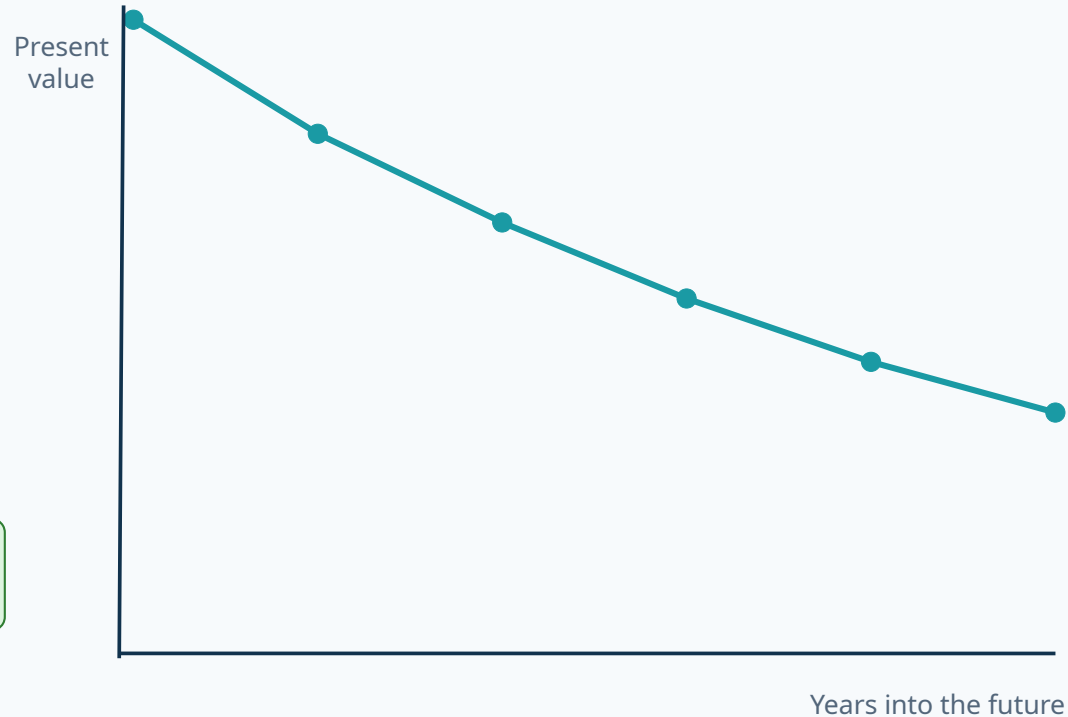
$$\text{Present value} = \text{Future value} / (1 + r)^t$$

r = discount rate • t = years in the future

Costs now



Health gains later

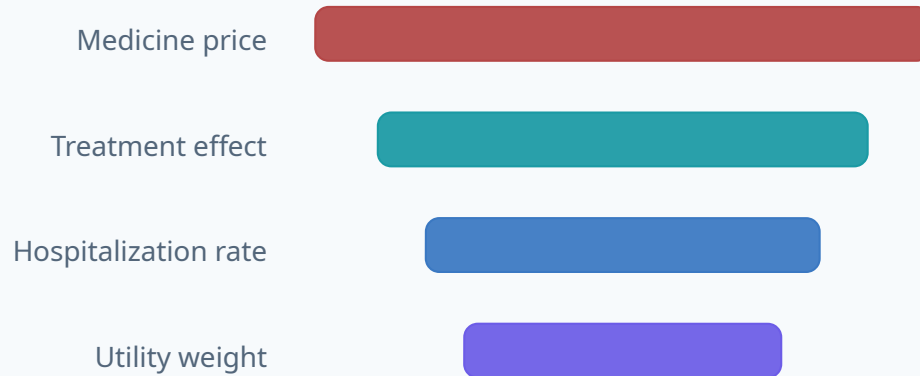


Higher discount rates reduce the present value of distant benefits, which can make vaccination, screening, and health-promotion programs appear less cost-effective.

Uncertainty: never stop at one ICER

Economic evaluations rely on uncertain evidence and assumptions.

DETERMINISTIC SENSITIVITY ANALYSIS



Tornado diagram: which inputs move the result most?

A single ICER can create false precision.

PROBABILISTIC SENSITIVITY ANALYSIS

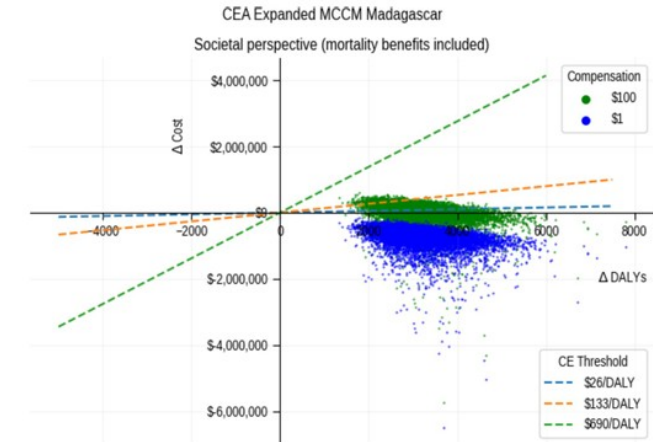


Fig. 1 Cost-effectiveness analysis plane from a societal perspective comparing the impact of nominal compensation of \$1 for CHWs versus \$100 per month.

Each point is one sampled result

DECISION UNCERTAINTY

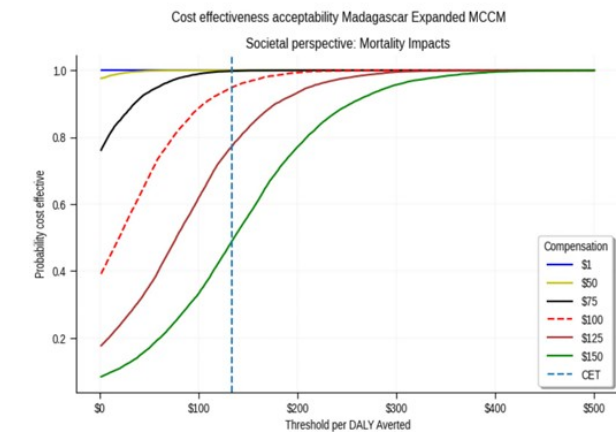


Fig. 2 Cost effectiveness acceptability curve with willingness-to-pay thresholds on the x-axis. The vertical dashed line represents a threshold of \$133 per DALY averted.

Probability cost-effective rises with the threshold

Net monetary benefit simplifies the decision rule

Convert health gains into a monetary-equivalent value using the threshold.

$$\text{Incremental NMB} = \lambda \times \Delta\text{Effect} - \Delta\text{Cost}$$

λ = the threshold value placed on an additional unit of health

$$\lambda = \$50,000/\text{QALY}$$

$$\Delta\text{Effect} = 0.2$$

$$\Delta\text{Cost} = \$600$$



$$\text{NMB} = \$9,400$$

Positive incremental NMB favors the new intervention.

Important: NMB is not financial profit. It is a mathematical transformation of health benefits using the threshold.

Beyond the ICER: equity, transferability, implementation

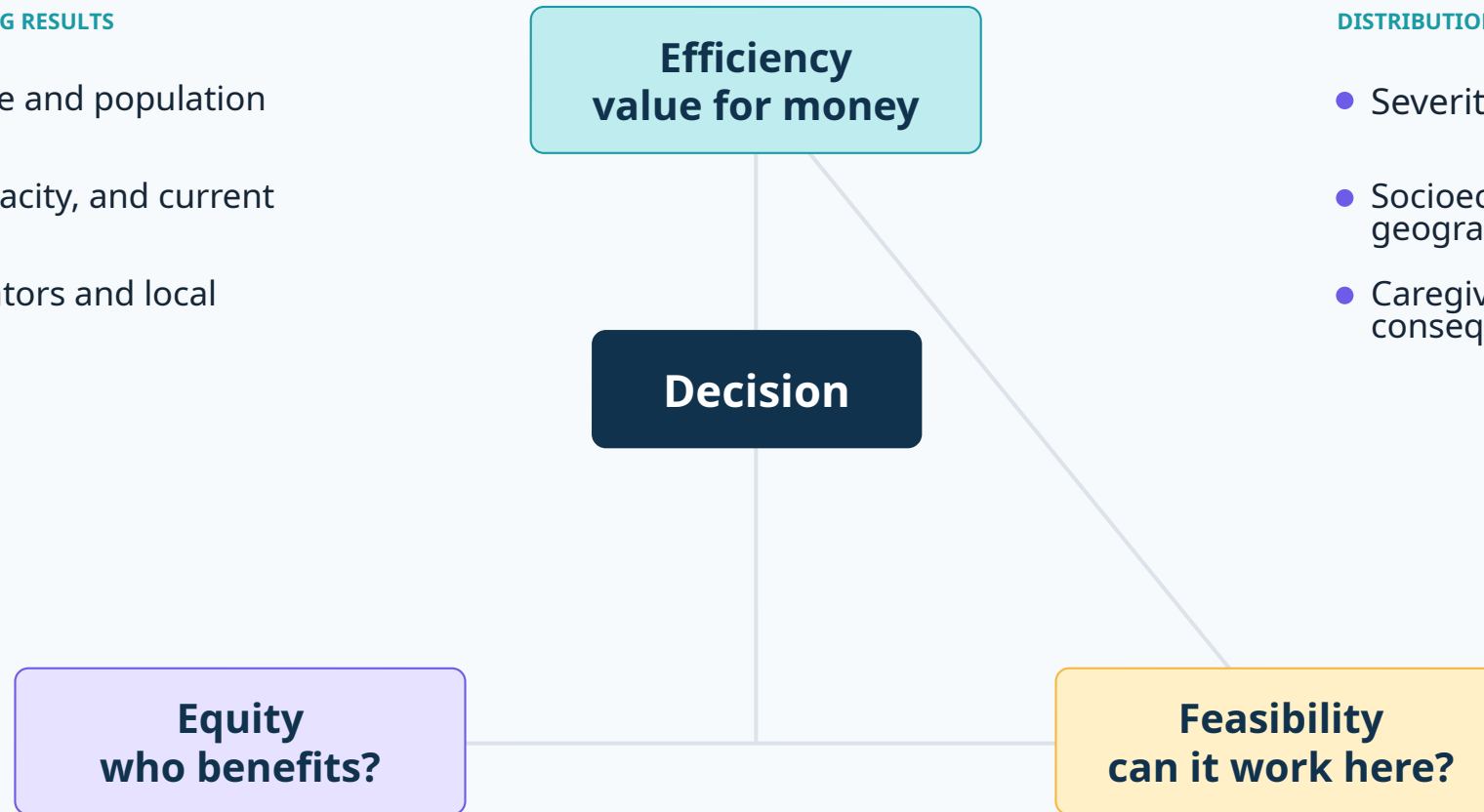
Economic evaluation is a decision aid, not a decision-making machine.

CHECK BEFORE TRANSFERRING RESULTS

- Disease prevalence and population characteristics
- Prices, wages, capacity, and current practice
- Available comparators and local thresholds

DISTRIBUTIONAL QUESTIONS

- Severity and unmet need
- Socioeconomic and geographic inequality
- Caregiver and household consequences



A favorable ICER may still fail if labs, referral services, workforce, or patient access are missing.

Critical appraisal: what to ask of any study

A mathematically correct ICER can still be misleading.

- 1 Question**
Was the decision problem well defined?
- 2 Alternatives**
Were relevant comparators included?
- 3 Evidence**
Was effectiveness credible and applicable?
- 4 Costs + outcomes**
Were important categories identified, measured, and valued?
- 5 Timing**
Were time horizon and discounting appropriate?
- 6 Incremental**
Was analysis based on additional cost and benefit?
- 7 Uncertainty**
Were sensitivity and scenario analyses transparent?
- 8 Interpretation**
Were affordability, equity, feasibility, and context discussed?

Bottom line: economic evaluation makes trade-offs visible so limited resources can generate greater health while accounting for affordability, feasibility, equity, and population need.

Principal reference: Drummond MF, Sculpher MJ, Claxton K, Stoddart GL, Torrance GW. Methods for the Economic Evaluation of Health Care Programmes. 4th ed. Oxford University Press; 2015.